Lesson 9-2 · Session 1

# From Table to Graph

Name _____ Date _____ Period _____

Maryland Standard 6.AT.11 · Learning target — I can turn a table of values into points and plot them on a graph.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

## Learn from a model

**REMEMBER**

- Each row of a table of values becomes one ordered pair (x, y) to plot on a coordinate graph.
- Plot several points from the table, then look for a pattern (often a straight line).
- Watch out — a fee shifts the line up so it no longer starts at the origin.
- Check it — every plotted point should satisfy the rule from the table.

**WORKED EXAMPLE**

Turning a Table into Points on a Graph A rental site charges $7 per movie. Make a table for 1, 2, and 3 movies, then list the points to plot.

**Step 1** Independent: number of movies (x). Dependent: total cost (y).

**Step 2** Multiply each x by the rate: 1 7 · 2 14 · 3 21

**Step 3** Write each row as an ordered pair (x, y): (1, 7), (2, 14), (3, 21).

**Step 4** Plot each pair: move right along the x-axis first, then up to the y value.

**Answer Points to plot: (1, 7), (2, 14), and (3, 21). They rise in a straight path.**

## Practice

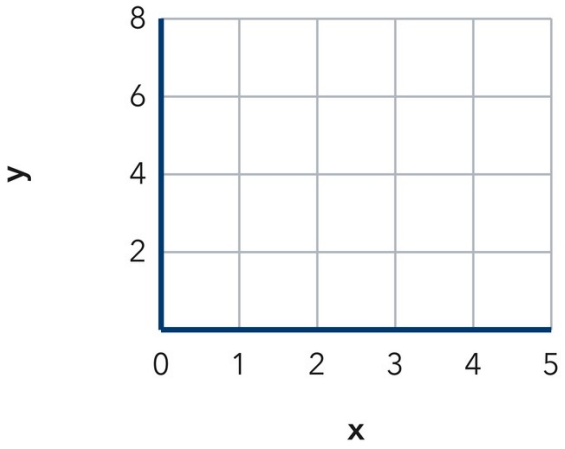
**1.** Using the table x: 1,2,3 y: 2,4,6 — write the three ordered pairs. _____

Answer _____

**SHOW YOUR THINKING**

**2.** Plot the three ordered pairs from problem 1 on a coordinate plane.

Graph. Plot the three ordered pairs and draw the line through them and the origin.




Answer _____

SHOW YOUR THINKING

**3.** Do the points in problem 1 form a straight line? What does this tell you about the relationship?

Answer _____

SHOW YOUR THINKING

**4.** Write each table row as an ordered pair (x, y).

a. x = 3 and y = 15 _____
b. x = 6 and y = 12 _____
c. x = 0 and y = 0 _____
d. x = 5 and y = 45 _____
e. x = 10 and y = 70 _____

Answer _____

SHOW YOUR THINKING

**5.** Trail Mix. A corner shop sells trail mix for $4 a pound. Make a table for 2, 3, and 6 pounds, then write the three ordered pairs.

Answer _____

**SHOW YOUR THINKING**

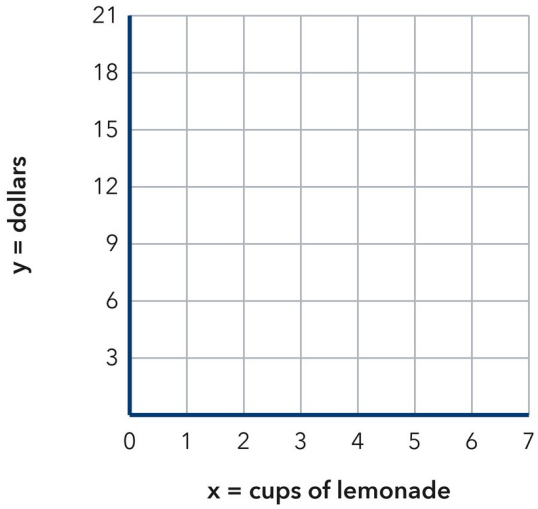
**6.** Stair Climb. Kwame climbs 14 stairs for every floor of his apartment building. Write the ordered pairs for 2, 5, and 8 floors, then explain what the middle point means.

Answer _____

**EXPLAIN WITH MATHEMATICAL EVIDENCE**

**7.** Lemonade Stand. A stand charges $3 for each cup of lemonade. Plot the points for 1, 2, 4, and 6 cups, label each axis with its quantity, then tell whether the points lie on a straight path.

Graph. Plot the four points, label each one, and draw the line through the origin.




Answer _____

**SHOW YOUR THINKING**

**8.** A student plotted the pairs from the table x: 2, 4, 6 and y: 10, 20, 30 by using each x-value vertically and each y-value horizontally. Explain the mix-up, then describe how to plot (4, 20).

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

**9.** At $6 per book, list the ordered pairs for 1, 2, and 4 books.

Answer _____ three pairs

SHOW YOUR THINKING

**10.** A table gives (1, 11), (2, 22), and (3, 33). Describe the rule and the shape of the graph.

Answer _____ rule and shape

RECORD YOUR RESPONSE

**11.** With a Fee. A plan charges $10 plus $4 per month. List the pairs for 1, 2, and 3 months.

Answer _____ three pairs

SHOW YOUR THINKING

**12.** Find the Mistake. For x = 2 and y = 16, Cass plotted (16, 2). Explain the mistake.

Answer _____ the mistake

EXPLAIN WITH MATHEMATICAL EVIDENCE

## Spiral review

**13.** From Lesson 9-1 Session 2. A rule is “multiply by 6, then add 2.” Find y when x = 5.

Answer _____

SHOW YOUR THINKING

**14.** From Lesson 8-2 Session 1. Solve x + 34 = 61.

Answer _____

SHOW YOUR THINKING

**15.** From Lesson 6-6 Session 1. Simplify 6n + 7 + 2n.

Answer _____

SHOW YOUR THINKING

## Test practice

**16.** A row of x = 7, y = 35 becomes —

| | |
|---|---|
| □ A. (42, 0) | □ B. (35, 7) |
| □ C. (7, 7) | □ D. (7, 35) |

**17.** A graph with a starting fee does not pass through the —

- □ A. origin
- □ B. x-axis
- □ C. y-axis
- □ D. first quadrant

**18.** At $12 per hour, 4 hours gives the pair —

| | |
|---|---|
| □ A. (48, 4) | □ B. (4, 48) |
| □ C. (4, 12) | □ D. (12, 4) |

## Check and revise

| CHECK YOUR MATHEMATICS | HOW DID IT GO? |
|---|---|
| Check every pair using the same rule and label quantities and units.<br>Every part answered / Work shown / Units labeled when needed | I can do this:<br>Independently / With a hint / With help<br>Question to revisit: _____ |

One correction or question I want to discuss: _____

Substitute one input-output pair into your rule. Check units and whether your answer fits the situation.